Team Control Number

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Summary

The NBA is a professional basketball league based in North America and is one of the largest sports leagues in the world. Balancing both entertainment and management, NBA team owners and general managers constantly seek opportunities to improve their teams' performance and profits. Although both of these metrics are composed of many factors, we focused on the areas with the most potential for large, impactful benefits. For the former, this maps to the careful acquisition and movement of players based on predicted gains. For the latter, the precise pricing of tickets and other sold goods can significantly increase the potential revenue of a team, which improves profit margins.

First, we consider the offseason. In terms of performance, we see two main areas to improve. The first has to do with the draft; we use an **impact regression model** to assess the top college and international prospects by NBA standards, allowing for sifting out skilled players among a large draft. We also consider free agents, using a **team-value regression model** that predicts how active NBA players would perform on any given team, enabling us to rank these potential free agents by value.

Using similar approaches, we also show that revenue can be optimized by removing players with high salaries relative to their actual team impact; these players would likely be cut or traded to another team. For the season itself, we occasionally see opportunities to raise team performance, but consistently have opportunities to increase revenue. By formulating and solving a **constrained nonlinear optimization problem**, we see significant improvements to revenue by optimizing the ticket prices to games, significantly improving outcomes for the team's general manager and owner. We also analyze the cost that every team incurs by using the optimal prices for revenue. By doing this, we can also evaluate all teams' net profit. The season also often involves trades, which are performed through our **team-value regression model** to find consistent opportunities for trades by accurately predicting player values to both teams.

Through the use of our models, we expect that owners and general managers will consistently see improvements to their teams' chances of winning and overall profit. Through statistically assessing the performance of large sets of players and optimizing the prices of tickets in relation to consumer demand, our models will allow for streamlining previously unrefined or vague selections and values.

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1 Introduction

1.1 Background Information

The NBA is one of the most popular sports leagues in the world. As such, the teams that make up the league are both hugely profitable, and extremely difficult to manage. In this suggestion to the owner and general manager, we discuss ways to improve both winning and revenue.

1.1.1 Trades

Trades are events in which a group of players and/or draft picks from one team are exchanged with another team or teams for other players and/or draft picks. These range from simple trades where one player gets traded for another player on different teams, to complex trades involving many different teams, players, and draft picks.

In general, trades occur when player(s) or their team aren't happy with the other and/or because some player(s) would be more valuable to the team receiving them than the team they're currently on, and the same is true for some player(s) on the other team. That way both teams can become better overall from the trade.

1.1.2 Signings, Waivers, and Free Agency

Signings: Teams under the salary cap can sign free agents, subject to limitations. Teams over the cap use exceptions (e.g., Mid-Level Exception) to sign players.

Waivers: When a team waives a player (releases them) during the season, they are placed on the waiver wire for 48 hours. Other teams can claim the player by absorbing their existing contract. Priority goes to teams with the lowest winning percentage. If no team claims the player within 48 hours, they clear waivers and become an unrestricted free agent, allowed to sign with any team. If a guaranteed contract is waived, the original team still carries that salary against their cap, sometimes stretching it over many years.

Free Agency: There are two types of free agents: unrestricted and restricted free agents. A player whose contract expires becomes an unrestricted free agent who can sign with any team. A restricted free agent's team can match other teams' offers to keep their player. A team can also sign a player and immediately trade them to another team in a sign-and-trade.

1.1.3 Drafts

In the NBA, there is an annual event called The NBA Draft. In this event, teams have the option to **draft** players not currently in the league onto their

roster. This event is very important for the NBA, as it allows both organized acquisition of young talent, and a balancing system for especially successful or unsuccessful teams.

The draft structure is as follows: All the teams that miss the playoffs (14 total) are entered into a lottery. First, the #1 overall pick is drawn, which has a 14% chance to go to the bottom 3 teams in terms of record, and odds decrease incrementally for all other teams until the team with the best record has a 0.5% chance. The #2, #3, and #4 picks are drawn similarly, but if a team that has been drawn in an earlier pick is drawn again, that draw is discarded and redone. After the top 4 picks are drawn, the other 10 teams are given picks based on regular season record, with the team with the worst record given the #5 pick, and so on until the #14 pick [1].

After the first 14 picks, the 16 remaining teams are given picks in order of their regular season record, with the worst record out of those teams being given the #15 pick and so on. Finally, the second round of the draft (picks 31-60) follows the same order as the first round. Over the entire draft, if any teams are tied in their regular season record, a coin is flipped to determine the teams' records' ordering [8].

Note that teams can trade away their draft picks for future years in exchange for players or other draft picks. If that's the case, the pick is treated the same as if it had been of the original team (i.e it is decided by the original team's record), but is handled by the team it was traded to [8].

1.1.4 Profitability and Economics

The average NBA franchise is worth $5.51 billion [16], and has a revenue of $416.6 million [14]. With large numbers such as these, any marginal change in business can lead to a very large difference in total profit. The revenue breakdown of the average NBA team is as such: 38% from central revenue, meaning nationally-generated income for the entire league, which is distributed among the teams; 28% from stadium tickets; 14% from team sponsorships; 10% from local media; and 9% from miscellaneous sources [15].

For individual teams, the sources of revenue that can be manipulated are:

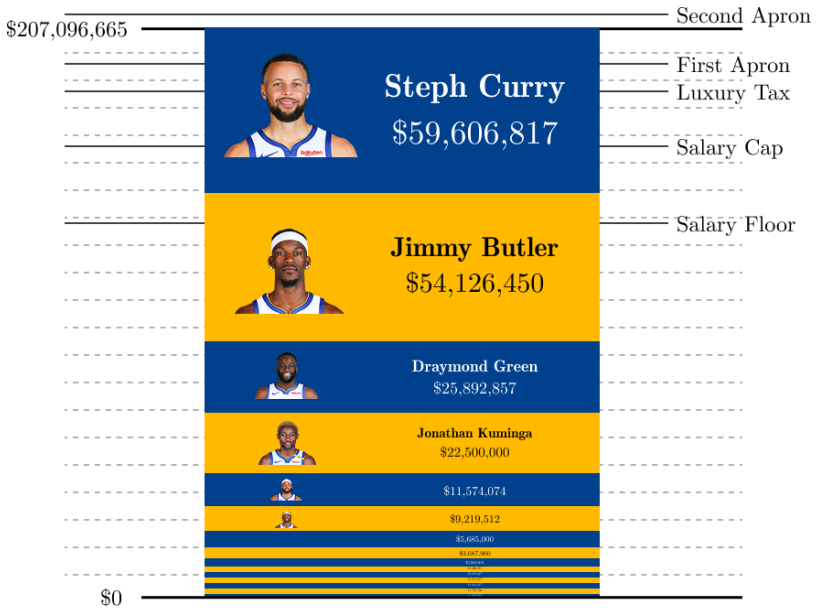
- Ticket Pricing: Each team can control all the numerous different prices for different types of tickets.
- Team Sponsorships: Teams sign sponsorship contracts with many different organizations, in which the teams advertise said sponsors. Teams can negotiate the pricing once these contracts expire.
- Local Media: Many teams sign television rights contracts for their games, and they can negotiate those contracts once they expire.

Other sources of revenue that are not set by the individual teams are:

- Central Revenue: While the league itself receives some of the earnings of the teams, it redistributes a chunk of that income to the teams themselves. While the numbers are not disclosed, it is estimated that if a team contributed to some earnings (i.e a player of that team had their jersey sold), they may earn a larger fraction than other teams [15].
- Miscellaneous sources: NBA teams earn about 9% of their cash from sources such as prize money, stadium parking, stadium food & drinks, and others [15]. These prices are often set by the individual businesses, and cannot be manipulated by the team itself.

The major expenses of each NBA team and owner are:

- Salaries: Salaries are one of the biggest expenses of most NBA franchises. They are simply the sum of the yearly salaries of every player. The average NBA franchise spends $196.7 million annually on player salaries [13].
- Luxury Tax: The luxury tax is a tax that NBA franchise **owners** are taxed by the NBA if the sum of their players' salaries exceeds the luxury tax level, which is $187.895 million in the 2025-26 NBA season [17]. The ratio between the amount an owner is taxed and the amount the franchise is over the luxury tax cap increases every $5 million that the team is over the cap. In addition, teams that are consistently above the luxury tax cap are taxed more per dollar over the luxury tax cap. [17]. There are 16 owners in the league who don't pay luxury tax at all, and the highest paying owner is the Minnesota Timberwolves owner who is taxed $119.6 million annually in luxury tax.
- Transportation: NBA players need to be transported to and from many games every season. In 2010, the New York Knicks published the amounts the franchise spends on necessities such as transportation, nutrition, etc. They reported expenses of $2.4 million, which would be $3.55 million adjusted for inflation [9].
- Nutrition: The Knicks additionally reported $4.59 million spent on player nutrition, including chefs, foods, and team-funded restaurants [9]. NBA teams are also required to allot $156 for every player per away day to spend on food [10].
- Arena and Training Facilities Maintenance: Staples Center quotes $10,000 per home game for maintenance, plus $1,000,000 a year for expansion and repairs, plus $10,000,000 every decade for expansions and such. That all totals to roughly $2,800,000 every year.
- Coaching Staff: While numbers aren't publicly available, the average head coach earns $12 million, and other primary assistants earn around $1 million per year, for about 5 coaches. In addition, there are about 20 coaches lower down who earn $200,000. All in all, the coaching staff costs about $21 million annually.



2025-2026 Warriors roster, sorted by salary

1.2 Problem Restatement

For this modeling problem, we were tasked with analyzing data for a sports team of some professional sport with more than 5 people on that team. To start, we were tasked with creating a dynamic decision-making model for the general manager (GM) and owner of our chosen team. This model should include both financial and sports decisions. This model should account for different situations, and aim to optimize revenue and sports performance.

Our model should include strategies for drafts, trades, and free agency signings. Additionally, we need to account for overall player value, player value specific to our team, and the profit the player might yield to the team and its owner.

Our model is generalized to all NBA teams, but in particular all letters and specific decisions are specific to the Golden State Warriors.

2 Terminology and Assumptions

2.1 Terms

- Plus/Minus (PM): For a given player, the change in relative points for the duration they are on for. Positive if the team gains points, negative if the other team gains points. Used for calculating the impact score.

- Minutes Played (Mins): The amount of minutes a player is on for during a specific game. Minutes played is not necessarily contiguous. Used for calculating the impact score.
- Impact Score (Impact): A metric for comparing the amount players will contribute to a team winning, irrespective of the team's environment. Derived as the product of the plus/minus and minutes played for a player.
- Team Fit: The relative performance that a player will have on a team based on that team's statistics. Only assesses a player's change in performance based on being on a certain team, rather than their overall performance.
- Value Score (Value): A metric for assessing the performance a player will have on a given team, based on both that team's environment and the overall performance of a player. Calculated as the product of the impact score and team fit.

2.2 Assumptions

- The contribution of players sum linearly to the overall contribution of a team, with regard to both winning (in impact and value) and profit (in salaries and profit).
- While we plan to make recommendations for the Golden State Warriors, our model is built to account for all teams. Thus, we assume that all other teams follow the same logic as expressed in our model. For example, we would only propose a trade if the trade is worthwhile for both the Warriors and the other team according to our model.
- Player popularity is accurately reflected by both social media following and all-star votes.
- Player picks during the draft will be among the top 100 prospects, rather than also considering less skilled players.

2.3 Justifications

- In our model, we account for players fitting differently on different teams. Our impact statistic is built to be proportional to total team contribution for that player. Thus adding the impact statistics for all the players should be proportional to team performance. In addition, the amount of money spent on salaries is clearly a sum of the individual players' salaries. Finally, the primary earnings that stem from individual players are merchandise and team popularity, leading to a larger fan base. We assume that merchandise is sold per player, meaning a linear sum is justified. Player

contributions and popularity are both added linearly, so we assume total impact would also be linear.

- Our model is designed and capable of expressing choices for all other teams. Assuming our model is somewhat optimal, other teams might consider using the same. This assumption also simplifies possible draft offers, as we can determine whether the draft is also beneficial to the other team.
- While player "popularity" is a difficult statistic to measure, we would like it to be linearly correlated to total profit brought in by that player to their team. We believe that social media following (specifically Instagram following) accurately represents how many people "like" that person and might buy more tickets to watch them play. In addition, All Star Votes represent how talented people believe that player to be, which is again linearly correlated to the increase in the team's profit.
- During the 60-player draft, some teams may have preferences or internal rankings that might have people in a different order than the standard ESPN prospect ranking. However, it is very rare that a team would draft someone below the top 100. Hence we choose to exclude such players from our model.

3 Performance

3.1 Draft

3.1.1 Model

To quantify a player's projected impact to assisting with a win, we define an impact score based on two metrics: normalized plus/minus (PM) and average minutes played. These variables account for both on-court effectiveness and overall reputation.

As we are looking at a draft of college basketball players, we aim to assess these players based on NBA levels of skills. For each NBA position, we model both PM and minutes played as linear functions of player performance variables,

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where $\hat{y}$ represents either normalized PM or normalized minutes; $x_1, x_2, \ldots, x_n$ are performance statistics for a player; $\beta_0, \beta_1, \ldots, \beta_n$ are unknown coefficients.

The specific performance statistics are detailed below.

Param	Name	Description
x_1	BLK	# Blocks
x_2	STL	# Steals
x_3	%FG	% Baskets made
x_4	%3P	% 3-pts made
x_5	%FT	% Free throws made
x_6	TRB	# Rebounds
x_7	PF	# Fouls

The coefficients are determined using linear regression by minimizing the sum of squared errors. For a single observation the squared error is $(y_i - \hat{y}_i)^2$, and thus the optimal coefficient vector satisfies,

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

We ran this process across a large NBA games database containing around a million samples [2]. This directly gives a set of weights that quantify the influence of each performance feature on both PM and minutes for each position, as shown in the table below.

Position	Stat	BLK	STL	%FG	%3P	%FT	TRB	PF
Forward	**PM**	0.04819	0.03641	0.15839	0.02011	0.04156	0.05819	-0.06335
Forward	**Mins**	0.23783	0.31263	0.52119	0.06906	0.48588	0.73086	0.25600
Guard	**PM**	0.02036	0.03246	0.18610	0.01807	0.04536	0.08907	-0.06530
Guard	**Mins**	0.29792	0.30412	0.81920	0.04307	0.59559	1.14288	0.36468
Center	**PM**	0.05824	0.03823	0.14434	0.02043	0.05008	0.05078	-0.05794
Center	**Mins**	0.27761	0.27414	0.52827	0.04241	0.38761	0.80030	0.27343

Each of these features is scaled through basic normalization, using a min and max across the entire dataset independently of position,

$$x_{i,j} = \frac{x_{i,j} - \min_{1 \le t \le n} x_{t,j}}{\max_{1 \le t \le n} x_{t,j} - \min_{1 \le t \le n} x_{t,j}}$$

Using these position-specific weights, we evaluate a set of drafted players. The list of draft prospects [4] was sourced from ESPN, and performance data for these players was derived from NCAA [5] and international leagues [6].

For each drafted player, we compute the predicted values $\hat{y}$ using the regression models, giving normalized PM and minutes. These outputs are subsequently normalized across the drafted player pool.

The impact score for each player is defined as the product of their normalized PM and normalized minutes played,

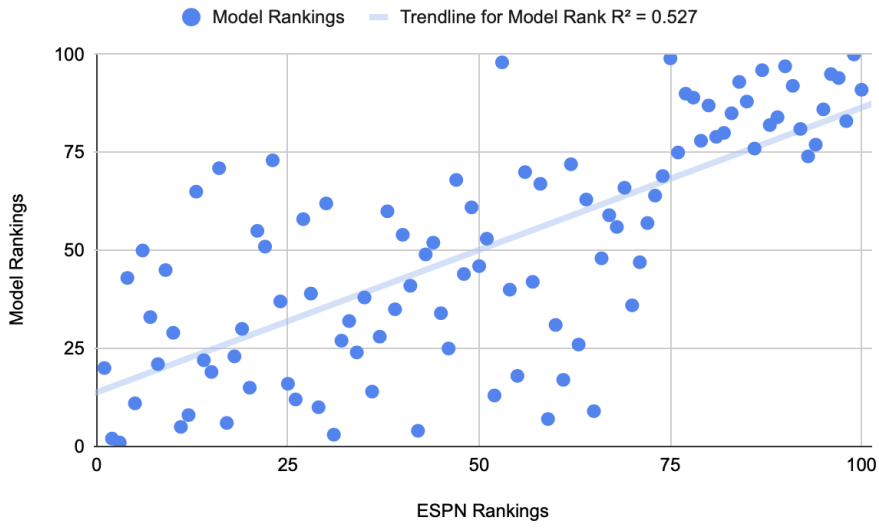
$$\text{Impact} = \hat{\text{PM}}_{\text{norm}} \times \hat{\text{mins}}_{\text{norm}}$$

Players are ranked by this impact score, allowing us to find players in the draft with the highest projected impact to winning. A set of the top ranked players in the draft is compiled in the table below.

N	Player	$\hat{\text{PM}}_{\text{norm}}$	$\hat{\text{mins}}_{\text{norm}}$	Impact
1	Bruce Thornton	1.00000	0.94373	0.94373
2	Cameron Boozer	0.99634	0.89253	0.88926
3	Tyler Tanner	0.92961	0.91604	0.85157
4	Caleb Wilson	0.93693	0.88939	0.83330
5	K.J. Lewis	0.81644	1.00000	0.81644
6	Blue Cain	0.89048	0.87479	0.77899
7	Silas DeMary Jr.	0.84262	0.91440	0.77049
8	Kingston Flemings	0.91328	0.83427	0.76193
9	Darryn Peterson	0.93271	0.81436	0.75956

3.1.2 Error

Comparing ESPN's top 100 prospects [4] and our results, we see a relationship between the purely statistical model rankings and the human-assessed ESPN rankings, with a R^2 of 0.527.



ESPN rankings compared to the model's rankings

We performed further analysis through finding squared error, based on taking $(\hat{y}_i - y_i)^2$ for each data point. The results are shown below.

Position	Stat	$\mathbf{RMSE_{norm}}$
Forward	**PM**	0.06175
Forward	**Mins**	0.08711
Guard	**PM**	0.06431
Guard	**Mins**	0.10091
Center	**PM**	0.06645
Center	**Mins**	0.08559

These RMSE values all range around the $0.06 - 0.10$ boundary, which is acceptable for this model based on the error having little impact to the selection of top players.

3.2 Trades

3.2.1 Model

For trades, we are aiming to not only consider the impact of players in general, but also how they will perform on our own team. This metric is quantified as team fit, where a higher team fit determines how much of a player's impact will translate to their actual performance. For each given player i, we first model their predicted plus/minus (PM) and minutes played if they played on a given team based on that team's statistics,

$$\hat{y}_i = \beta_{i_0} + \beta_{i_1} x_1 + \beta_{i_2} x_2 + \cdots + \beta_{i_n} x_n$$

$\hat{y}_i$ is either normalized PM or normalized minutes for the player i; $x_1, x_2, \ldots, x_n$ are statistics for the target team; $\beta_{i_0}, \beta_{i_1}, \ldots, \beta_{i_n}$ are learned coefficients.

The specific team statistics used are detailed below.

Param	Name	Description
x_1	BLK	# Blocks
x_2	STL	# Steals
x_3	%FG	% Baskets made
x_4	%3P	% 3-pts made
x_5	%FT	% Free throws made
x_6	TRB	# Rebounds
x_7	PF	# Fouls
$x_{8...11}$	Q1P-Q4P	% Quarter points
x_{12}	%BP	% Bench points
x_{13}	%FP	% Fast break points
x_{14}	%TP	% Points from turnovers
x_{15}	%PP	% Points in the paint
x_{16}	%SP	% Second chance points

To find the β_i coefficients, we regress over a dataset of every game played for a given player i, using the x statistics for the team they played with [2]. Using the predicted result, we compute squared error and minimize it to find the optimal β_i coefficients,

$$\hat{\beta}_i = \arg\min_{\beta_i} \sum_{j=1}^{n} (y_{i,j} - \hat{y}_{i,j})^2$$

Based on these models, we now express the team fit score based on a given player's projected PM and minutes played on that team,

$$\text{Team Fit} = \text{PM}_{\text{proj}} \times \text{mins}_{\text{proj}}$$

PM and minutes are not normalized in this equation to allow for not having players with predictions that are completely maximized for a given team. These statistics are normalized during the training process, however, so the fit is still able to be computed based on these values being centered around $0 - 1$.

For finding eligible players for a trade, we used a dataset of the 518 players playing in the 2026 season [3]. We then predicted the PM and minutes for these players based on the averaged team statistics of the Golden State Warriors from October 2025 to January 2026, representing the current stats [2]. We then compute the team fit scores and find the top players, summarized in the table below.

N	**Player**	**Team Fit**
1	Miles Kelly	1.10239
2	Oscar Tshiebwe	0.84496
3	Adou Thiero	0.63761
4	Kevin McCullar Jr.	0.57083
5	Jrue Holiday	0.46697
6	Curtis Jones	0.44138
7	P.J. Washington	0.43945
8	Daeqwon Plowden	0.42373
9	Ja Morant	0.41638

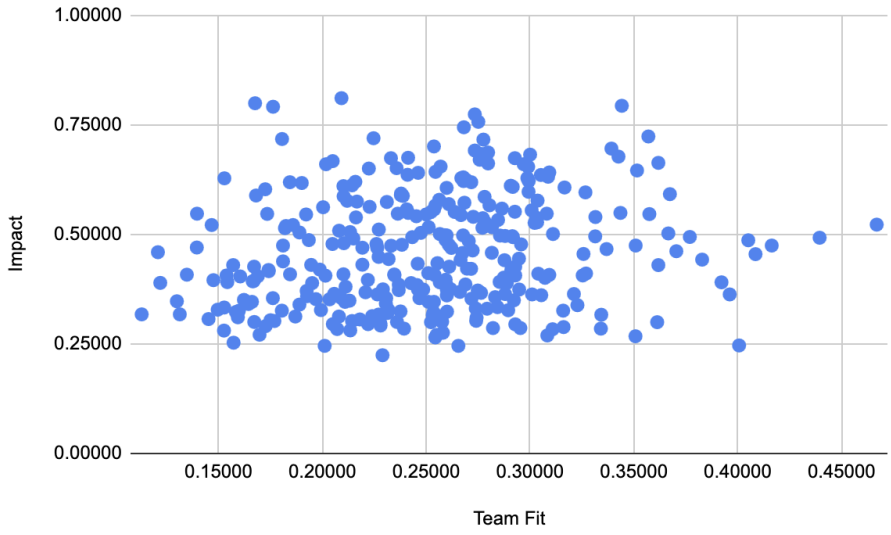
However, this list tended to bias towards players with low minutes played, as they will perform similarly on any team because of consistently hitting the same amount of minutes. Thus, a cutoff is added in # games ≥ 5 and average minutes played ≥ 15, following standard NBA leaderboard procedure. This creates a new table which accurately describes players who regularly receive play-time on their respective teams.

N	Player	Team Fit
1	Jrue Holiday	0.46697
2	P.J. Washington	0.43945
3	Ja Morant	0.41638
4	Kyshawn George	0.40863
5	Draymond Green	0.40512
6	Taylor Hendricks	0.40065
7	Bub Carrington	0.39621
8	Bilal Coulibaly	0.39220
9	Nic Claxton	0.38288

Based on these team fits, we now aim to find players that also have a high overall impact score without regard to team climates. We use the same impact formula as before, but now evaluate it based on the averaged normalized PM and minutes played for a given player for games they played from October 2025 to January 2026,

$$\text{Impact}_i = \text{PM}_{\text{norm}_i} \times \text{mins}_{\text{norm}_i}$$

Taking both the team fit score to the Warriors and independent impact score for each player currently in the league nets the following graph.



Team fit and impact for each player

As a final step, we now score each player on how valuable they are to the Warriors based on taking the product of these two statistics. We use the product

to provide weight to both projected impact and minutes on our team, and current impact on their own team. As a result, neither of our predictions could be too disjoint from reality, as they would be weighed down by the current impact.

$$\text{Value}_i = \text{Team Fit}_i \times \text{Impact}_i$$

Ranking by this statistic, we arrive at a final table which orders players by the total value that they give to the Golden State Warriors based on their team fit and impact scores.

N	Player	Value
1	Cade Cunningham	0.27343
2	Tyrese Maxey	0.25855
3	Jrue Holiday	0.24416
4	Austin Reaves	0.24016
5	Devin Booker	0.23623
6	Payton Pritchard	0.23229
7	De'Aaron Fox	0.22733
8	Giannis Antetokounmpo	0.21758
9	P.J. Washington	0.21666

3.2.2 Example

A prominent use case for computing the value of players to specific teams is finding potential trades that would increase the overall player value of one's team through presenting a trade that would be accepted. In this example, we calculated the values of other players to the Golden State Warriors and the values of Warriors players to other teams. Then, we found trades that would increase the values of both teams, which would present a benefit to both teams and thus be accepted. A few of the prominent trades are listed below, along with their increase to overall value for the Warriors (GSW) and the team being traded with (other).

Other	GSW Player	Other Player	GSW Gain	Other Gain
Grizzlies	De'Anthony Melton	Zach Edey	0.05095	0.07968
Spurs	De'Anthony Melton	Julian Champagnie	0.04272	0.01396
Lakers	Stephen Curry	Austin Reaves	0.03555	0.00180

Each of these trades is supported by our model, and would present opportunities for a general manager to predictably increase the value of the Golden State Warriors. These trades can also assist with lowering the total team salary to avoid reaching boundaries like the salary cap, luxury tax, and first/second aprons if the received player has a lower salary than the traded player.

4 Revenue

Profits are the difference between revenue and costs. One primary way in which we try to maximize profit is by optimizing the revenue of ticket sales. To determine the optimal average ticket price for every NBA team's arena, we construct an optimization model.

4.1 Parameters

To start, we define all of our parameters, which include constants, variables, and functions.

Param	**Description**
p_i^0	Current average ticket price of team i
cap_i	Arena capacity of team i
b_i	Constant of depreciation of demand for team i
a_i	Intercept of demand curve for team i
p_i^*	Optimal average ticket price of team i
$D_i(p_i)$	Demand curve for team i
$R_i^0(p_i)$	Current revenue function for team i
$R_i^*(p_i)$	Optimal revenue function for team i

The constant of depreciation b_i determines how sensitive demand is to price and is inversely proportional to the current ticket price:

$$b_i \propto \frac{1}{p_i^0}.$$

We calculate b_i as

$$b_i = \frac{\text{fractional decrease in demand}}{\text{fractional increase in price}} \cdot \frac{a_i}{p_i^0}.$$

For our use, we assume that an increase in price by 50% will lead to a 20% decrease in demand.

Our equation for b_i becomes

$$b_i = \frac{0.2}{0.5} \cdot \frac{a_i}{p_i^0} = 0.4\frac{a_i}{p_i^0}$$

4.2 Assumption for Demand

Because we are adjusting prices of tickets, we need to take into account that we likely won't have the same amount of people attending games if prices are

overly expensive. Thus, we need a decay curve to model our demand as opposed to assuming it is constant. We decided to model demand as

$$D_i(p_i) = a_i - b_i p_i$$

because as price, p_i, increases, demand should decrease. b_i acts as the slope of the demand function. It relies on the assumption we made earlier that a 50% increase in price, p_i, will lead to a 20% decrease in demand, D_i. This formulation of demand implies that demand is synonymous with the number of people that attend the game on average.

4.3 Formulation

The revenue for a team i can be calculated as the demand for that team multiplied by the average ticket price at that team's arena:

$$R_i(p_i) = p_i \cdot D_i(p_i) = a_i p_i - b_i p_i^2,$$

where $D_i(p_i) = a_i - b_i p_i$ is the linear demand function.

Each team maximizes its own revenue independently:

$$\max_{p_i} R_i(p_i)$$

subject to

$$0.8\text{cap}_i \leq D_i(p_i) \leq \text{cap}_i,$$
$$p_i \geq 0 \ \ \forall i$$

Essentially, each team chooses its ticket price p_i to maximize its own revenue, while ensuring that the number of tickets sold is between 80% of arena capacity and full capacity, and that prices are non-negative. The constraint comes from the NBA's Collective Bargaining Agreement (CBA) in which every team must sell at least 80% of the total arena seats on average throughout all of their season home games or incur a penalty. Thus, to not incur this fiscal penalty, we have made this a hard constraint.

4.4 Current Team Data

The current prices and respective ticket revenue values are:

Team	$\text{Price}_{\text{avg}}$	Attendance	Capacity	Revenue
Knicks	213.43	19799	19812	4225883
Lakers	146.20	18723	18997	2737242
Warriors	127.73	18064	18064	2307315
Celtics	114.18	19156	19600	2187232
Bulls	78.88	20138	20917	1588487
Wolves	73.13	18835	19000	1377429
Mavericks	64.00	20079	20100	1285078
⋮	⋮	⋮	⋮	⋮
Wizards	34.90	16188	18573	564947
Blazers	30.63	17546	19411	537421
Hornets	25.35	17169	19077	435238
Jazz	22.58	18175	18306	410392
Pelicans	23.28	16815	16867	391445
Grizzlies	22.00	16660	17794	366524

This shows the current average price, average attendance, and revenue per game for every NBA team.

4.5 Model Results

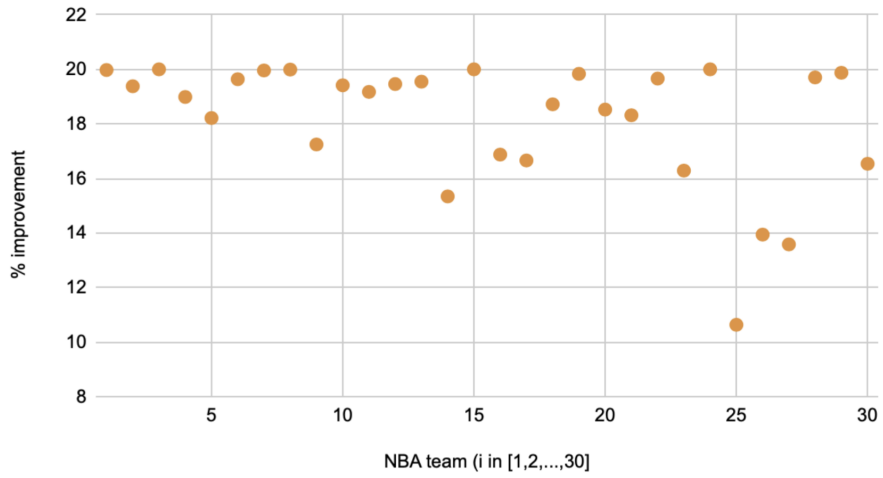
Next, we calculated the optimal price for each team for this optimization formulation. We found the best prices to maximize revenue for every NBA team.

Team	Price (p^*)	Demand	Revenue	Improvement
Knicks	319.88	15850	5070020	1.199754059
Lakers	215.01	15198	3267702	1.193793615
Warriors	191.59	14451	2768778	1.200000146
Celtics	165.98	15680	2602520	1.189869161
Bulls	112.22	16734	1877804	1.182133420
Wolves	108.42	15200	1647929	1.196380774
Mavericks	95.87	16080	1541563	1.199586733
⋮	⋮	⋮	⋮	⋮
Wizards	42.06	14858	625007	1.106311388
Blazers	39.43	15529	612331	1.139387336
Hornets	32.39	15262	494341	1.135795035
Jazz	33.54	14645	491252	1.197032590
Pelicans	34.77	13494	469240	1.198738980
Grizzlies	30.01	14235	427135	1.165367846

Optimal pricing results by team

4.6 Analysis

The leftmost column, improvement, is an indicator for how much better the optimal strategy is compared to the current. It is calculated by the optimal (max) revenue divided by the current revenue ($\frac{R^*}{R_0}$). We averaged out this data and determined that the optimal strategy yields a revenue 18.2% better than the current revenue on average. We also found the minimum and maximum improvement values. It is interesting to note that we found a minimum improvement percentage of 10% and a maximum improvement percentage of 20%. However, besides a few outliers, our optimal strategy improved revenue amounts by 16% to 19%:



This consistent and significant improvement shows that every team can drastically improve their revenue from ticket sales and thus dramatically improve profit. Ticket sale revenue is a major aspect of overall revenue: the Golden State Warriors currently receive a revenue of $\$2,307,315$ per game from their ticket sales. Annually, over their 41 home games, they make a total revenue of $\$94,599,904$.

Using our optimal model, if the Warriors increase their average ticket price from \$127.73 to \$191.59, we expect their per game revenue to be $\$2,768,778$ and their annual revenue from ticket sales alone to be $\$113,519,898$. This roughly 20% increase in revenue equates to almost a 20 million dollar increase in profits assuming that our demand curve is reasonable.

This shows how this completely achievable change can make a huge difference, as ticket sales are approximately 28% of the total revenue for every NBA team.

5 Extension to Profit

5.1 New Functions

We now have a working revenue model, but we want to model profit, so we also need to examine costs. First, we define our new variables:

Param	Description
$C_i^*(p_i)$	Costs of team i
γ_i	Implied marginal cost
$\Pi_i^*(p_i)$	Profit of team i

We define our cost function as

$$C_i^*(p_i) = \gamma_i \cdot D_i^*(p_i)$$

where D_i^* is our optimal demand function from before. Recall that demand also represents attendance, so we are essentially saying total cost = total people * cost per person, which makes intuitive sense.

Additionally, we know that total profit is the difference between total revenue and total cost, so we define profit as

$$\Pi_i^*(p_i) = R_i^*(p_i) - C_i^*(p_i)$$

So

$$\Pi_i^*(p_i) = p_i \cdot D_i^*(p_i) - \gamma_i \cdot D_i^*(p_i) = (p_i - \gamma_i)D_i^*(p_i)$$

However, we have a small problem: we do not know the values for γ_i.

5.2 Solving for γ_i

We start with our initial profit function $\Pi_i^*(p_i)$.

First, we can expand the profit function to get

$$\Pi_i^*(p_i) = (p_i - \gamma_i)(a_i - b_i p_i) = a_i p_i - b_i p_i^2 - a_i \gamma_i + \gamma_i b_i p_i$$

Grouping terms we get

$$\Pi_i^*(p_i) = -b_i p_i^2 + (a_i + \gamma_i b_i)p_i - a_i \gamma_i$$

Using the profit first-order derivative condition with the maximization rule, we know that

$$\frac{d\Pi_i^*}{dp_i} = \frac{dR_i^*}{dp_i} - \frac{dC_i^*}{dp_i} = 0$$

Plugging in our expanded and simplified profit function, we get

$$\frac{d\Pi_i^*}{dp_i} = -2b_i p_i + (a_i + \gamma_i b_i) = 0$$

So

$$2b_i p_i = a_i + \gamma_i b_i$$

Finally, we rearrange to isolate for γ_i

$$\gamma_i = \frac{2b_i p_i - a_i}{b_i}$$

However, since we want to find the cost function for the optimal revenue, γ_i should actually be

$$\gamma_i = \frac{2b_i p_i^* - a_i}{b_i}$$

We can also ensure that the first-order condition really provides us with the optimum by analyzing the second-order condition or concavity of the profit function $\Pi_i(p_i)$.

$$\frac{d^2\Pi_i^*}{dp_i^2} = -2b_i$$

Since $b_i > 0 \ \ \forall i$, $\frac{d^2\Pi_i}{dp_i^2} < 0$ so our solution for γ_i is a truly optimal value.

5.3 Cost Function

So we can now calculate the values for

$$C_i^*(p_i) = \gamma_i \cdot D_i^*(p_i)$$

γ_i	C_i
427.39	6,774,131.50
300.98	4,574,294.04
255.47	3,691,796.97
⋮	⋮
45.82	671,033.90
46.86	632,328.84
49.98	711,465.30

Values for γ_i and C_i based on optimal price p^* for revenue

5.4 Profit Function

Revisiting our profit function, $\Pi_i(p_i) = -b_i p_i^2 + (a_i + \gamma_i b_i)p_i - a_i\gamma_i$, we can now evaluate the profit for our optimal price for tickets, p_i^*.

$$\Pi_i(p_i^*) = -b_i(p_i^*)^2 + (a_i + \gamma_i b_i)p_i^* - a_i\gamma_i$$

Evaluating this, we get how much profit an NBA team i makes per game.

5.5 Final Results

Team	p_i^*	$R_i^*(p_i)$	$C_i(p_i^*)$	$\Pi_i(p_i^*)$	Improvement
Knicks	319.88	5,070,020	6,774,131.50	10,360,759.83	19.97540586
Lakers	215.01	3,267,702	4,574,294.04	6,819,162.07	19.37936154
Warriors	191.59	2,768,778	3,691,796.97	5,653,047.51	20.00001456
Celtics	165.98	2,602,520	3,746,579.20	5,501,526.91	18.98691610
Bulls	112.22	1,877,804	2,844,110.64	4,065,672.71	18.21334199
Wolves	108.42	1,647,929	2,261,912.00	3,408,404.70	19.63807737
Mavericks	95.87	1,541,563	2,062,420.80	3,152,097.58	19.95867334
Heat	94.74	1,494,640	1,994,401.92	3,052,615.73	19.99091849
Raptors	90.40	1,431,881	2,296,008.00	3,187,463.70	17.24416539
Magic	85.01	1,281,666	1,789,338.36	2,671,361.76	19.41476053
Nets	80.46	1,141,361	1,621,885.38	2,398,262.91	19.17218517
Thunder	76.78	1,118,135	1,555,804.08	2,327,019.37	19.45992939
Nuggets	66.45	1,063,216	1,469,600.00	2,206,033.50	19.54786739
Clippers	65.79	947,357	1,675,008.00	2,216,244.50	15.34015675
Cavaliers	62.85	977,041	1,302,754.80	1,994,791.96	20.00000491
Kings	66.45	936,227	1,532,178.75	2,105,370.58	16.87381490
Pistons	57.24	931,087	1,541,528.82	2,106,019.15	16.65714323
Spurs	62.43	916,675	1,344,081.82	1,954,474.46	18.71705046
Bucks	55.54	777,578	1,050,980.00	1,597,202.26	19.83421399
Rockets	53.23	768,821	1,141,653.76	1,649,016.66	18.52366639
Hawks	55.99	763,470	1,149,021.45	1,648,042.48	18.31557447
Sixers	47.03	752,752	1,031,426.64	1,555,697.05	19.66147811
Pacers	49.69	712,493	1,202,528.06	1,627,386.18	16.28530504
Suns	49.95	682,157	909,556.20	1,392,737.54	19.99997185
Wizards	42.06	625,007	1,342,866.04	1,627,140.24	10.63113876
Blazers	39.43	612,331	1,153,649.41	1,481,578.14	13.93873355
Hornets	32.39	494,341	945,786.14	1,206,081.34	13.57950346
Jazz	33.54	491,252	671,033.90	1,013,873.56	19.70325896
Pelicans	34.77	469,240	632,328.84	962,573.80	19.87389802
Grizzlies	30.01	427,135	711,465.30	969,076.71	16.53678456

Cumulative optimal prices, revenues, costs, profits, and percentage improvement from original to optimal values for all NBA teams per season for home games.

6 Code Descriptions

6.1 Impact Regression

We used two datasets, one containing data describing each individual player, and the other containing rows for every player in every game [2]. From the player data, we store the positions that each player is on; from the game statistics data, we derive features (blocks, free throw percentage, rebounds, etc) and labels (either plus/minus or the number of minutes played), accumulated by the player's position and the label type. We also find the global minimum and maximum values for each feature, which are used for normalization. We aim to create a model for every position, and both of the labels; thus, each model uses a different subset of the overall data. We loop through the position and label combinations, using scikit-learn's linear least-squares (OLS) solver to compute the regression weights. We now take the dataset we want to evaluate, which in this case is the list of the top 100 draft prospects [4]. From another dataset of NCAA basketball stats, we loop through each player, finding their features from a dataset [5] [6]. We run the models on each player using each label, and calculate it based on the player's position, giving impact scores.

6.2 Value Regression

We are trying to see how player performance changes based on team environments, so we start by collecting the games that each given player has played in [2]. At the same time, we also take the games that occurred in the past few months for calculating the average plus/minus and minutes played for each player, used later in the value calculation. Now, for each player, we again use scikit-learn to create a regression model that correlates chosen features that dictate the team environment with the labels of plus/minus and minutes. Through this, we can get the relative performance, or team fit, by using the model any given team's statistics. We used this model by assessing the value of each of the 518 NBA league players to the Golden State Warriors [3]. For each player, we evaluate team fit using the Warriors' statistics and take the product of it with previously computed impact score (itself the product of plus/minus and mins played) to produce an overall value score.

6.3 Convex Optimization

To find solutions for the revenue model, we wrote a script to solve the constrained convex optimization problem. Using numpy, we perform vectorized calculations based on our input data to find the domain sections that are within the constraints by clipping the domain of the model function based on the constraints. Finally, we evaluated the constrained region to find the optimal prices for every NBA team that would yield the maximum revenue.

7 Letter to Owner and General Manager

Dear Joe Lacob and Mike Dunleavy Jr.,

We have designed a continuous prediction algorithm to optimize both team performance and profits. Our algorithm is divided into 3 sections: Trades, Drafts, and Financial Concerns.

To start, we'd like to address the issue of the NBA draft. Our algorithm accurately predicts the performance of draft prospects in the NBA. Combined with some knowledge of which positions are currently most wanted on the team, we can propose a strategy for the next draft. Based on our model, we believe that Caleb Wilson, UNC, and Cameron Boozer, Duke, would be the best draft picks for the Warriors franchise in the 2026 draft. If both have already been selected, follow the algorithm and find the most valuable draft prospect still available.

Regarding trades, we have designed a model that determines how well any player in the league would do on the Warriors franchise, by weighing both overall performance and potential on the current Warriors team in particular. We have determined that the player that would be the best on our franchise would be Cade Cunningham, and the second and third places are Tyrese Maxey, and Jrue Holiday.

We have come up with some possible one-for-one trades. We have determined that one particular player, De'Anthony Melton, doesn't fit too well on our Warriors franchise. Thus, we concluded that trading him for Memphis Grizzlies center Zach Edey or San Antonio Spurs wing Julian Champagnie would both drastically improve our team performance. Another slightly less impactful but still useful one-for-one trade would be acquiring the young and promising Los Angeles Lakers guard Austin Reaves for Stephen Curry.

Addressing profit, we have designed a model to calculate revenue and cost for ticket sales and respective arena costs. After making an assumption about demand and arena attendance, we were able to model and find the optimal average price that would maximize total revenue. For the Warriors' home games at Chase Center, increasing the average ticket price from \$127.73 to \$191.59 would boost revenue substantially. Although this may only seem like a small change, this move would increase annual ticket sale revenue by over 19 million dollars, an over 20% increase from the status quo.

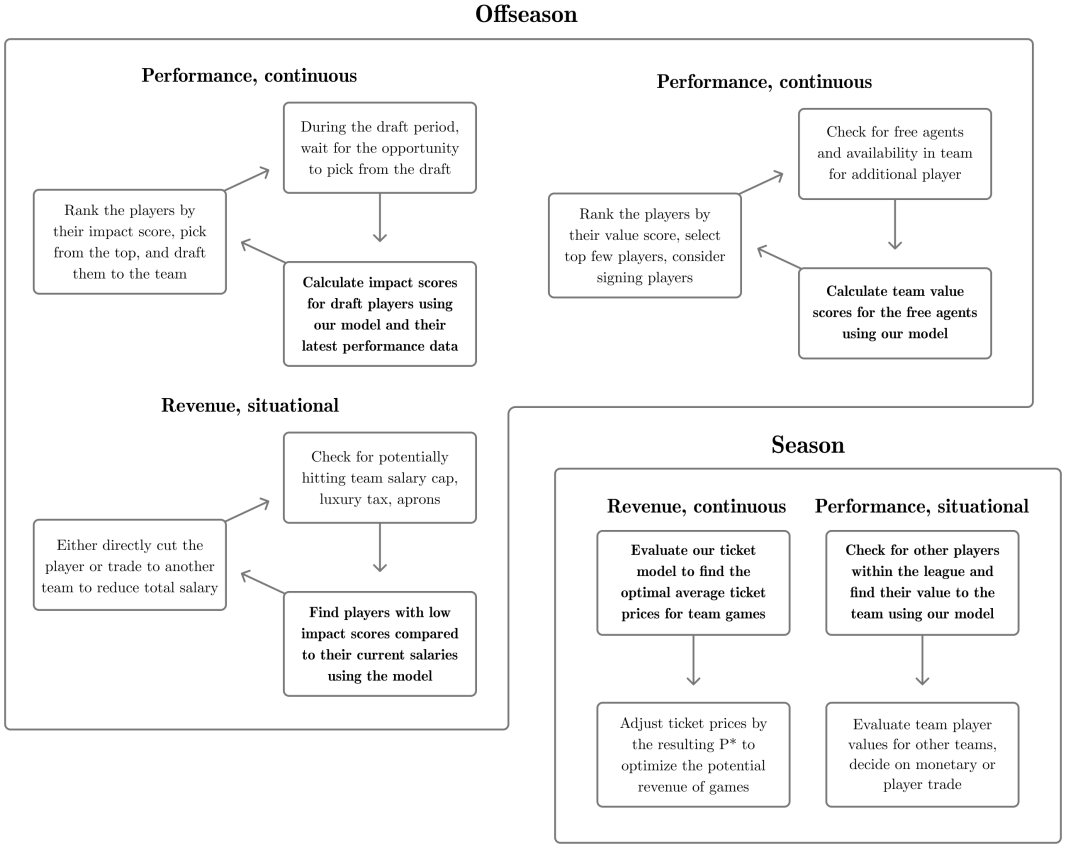
We also calculated the cost and profit for this maximized revenue and determined that this action would lead to a huge boost in net profit of close to 15 million dollars.

All of these moves, draft, trades, signings, ticket price, are designed to help the Golden State Warriors improve their basketball performance while also cutting down on costs and raising revenue, which should help the Warriors as they make a playoff push for this season and cut down on their expensive luxury tax, expenses, and payroll.

We created a flow chart that should help guide the decision-makers through offseason and midseason moves regarding both performance- and profit-based actions based on our solution. Note: the bolded sections are the parts we explicitly modeled and solved with code.

Sincerely,

Team #2631663



Guiding flow chart of decisions to make as general manager and/or owner.

8 References

[1] https://www.nba.com/news/nba-draft-lottery-explainer

[2] https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores

[3] https://www.basketball-reference.com/leagues/NBA_2026_advanced.html

[4] https://www.espn.com/nba/story/_/id/46886245/2026-nba-draft-big-board-rankings-top-100-prospects-players

[5] https://www.cbssports.com/college-basketball/stats/player/scoring/all-conf/all-pos/all/

[6] https://basketball.realgm.com/nba/stats

[7] https://www.basketball-reference.com/leagues/NBA_2026_advanced.html

[8] https://www.espn.com/nba/story/_/id/40388476/how-do-nba-draft-free-agency-work

[9] https://ivypanda.com/essays/the-cost-of-running-a-nba-basketball-team

[10] https://imgix.cosmicjs.com/25da5eb0-15eb-11ee-b5b3-fbd321202bdf-Final-2023-NBA-Collective-Bargaining-Agreement-6-28-23.pdf

[11] https://www.nba.com/news/the-athletic-2026-nba-draft-top-100-prospects

[12] https://sportsbusinessclassroom.com/nba-2025-26-luxury-tax-tracker/

[13] https://www.spotrac.com/nba/cap

[14] https://www.statista.com/statistics/193704/revenue-of-national-basketball-association-teams-in-2010/

[15] https://www.sportico.com/leagues/basketball/2025/how-nba-teams-owners-make-money-1234874170/

[16] https://www.sportico.com/feature/nba-team-values-ranking-list-1234697991/

[17] https://www.spotrac.com/nba/cap

9 Appendix

9.1 AI Appendix

1. Query: where can I find an accurate dataset on nba jersey sales?

   - Answer: There is no fully public, raw NBA jersey sales dataset. Best available sources:
     • NBA / NBPA "Top-Selling Jerseys" reports (official rankings, not unit sales)
     • Fanatics press releases (league's retail partner; partial data)
     • Statista / market research firms (paid, estimated figures)
     For exact sales numbers, data must be licensed directly from Fanatics or the NBA.

2. Query: latex how to format table

   - Answer: Basic LaTeX table format:

```
\begin{table}[h]
\centering
\begin{tabular}{l c r}
\hline
Column 1 & Column 2 & Column 3 \\
\hline
A & 10 & 0.5 \\
B & 20 & 1.2 \\
\hline
\end{tabular}
\caption{Example table}
\label{tab:example}
\end{table}
```

3. Query: python spawn multiple processes

   - Answer: Minimal example using ProcessPoolExecutor:

```
from concurrent.futures import ProcessPoolExecutor, as_completed
import os

def work(x):
    return x, os.getpid()

if __name__ == "__main__":
    with ProcessPoolExecutor(max_workers=4) as executor:
        futures = [executor.submit(work, i) for i in range(10)]
        for f in as_completed(futures):
            print(f.result())
```

Key points:
• Must be under if __name__ == "__main__":
• Functions must be top-level (picklable)
• max_workers controls number of processes

4. Query: what is the trb stat in an nba data table?

   • Answer: TRB = Total Rebounds.
   It is the sum of:
   • ORB (Offensive Rebounds)
   • DRB (Defensive Rebounds)
   Formula:

   ```
   TRB = ORB + DRB
   ```

   Used in NBA box scores and player/team stat tables.

5. Query: nba non-game score box dataset per game

   • Answer: An NBA Hustle Box dataset per game (player- or team-level hustle stats for each game) isn't directly available as a ready-made CSV publicly, but you can get it from these sources:
   Official NBA API / Stats Endpoints
   • The NBA's official Hustle Stats Boxscore API returns hustle stats by game (e.g., deflections, loose balls, charges, contested shots, screen assists) for players/teams. It's part of the NBA Stats API used on stats.nba.com.
   Third-party APIs Packages
   • balldontlie API v2 advanced stats includes hustle tracking data that can be queried per game and player.
   • hoopR (SportsDataverse) provides functions like nba_hustlestatsboxscore(game_id) that pull hustle stats as structured data frames (player + team hustle stats) from the NBA Stats API.
   • Other NBA API wrappers (e.g., nba_api in Python) expose hustlestatsboxscore endpoints.
   Official NBA Stats Portal
   • The NBA's Hustle Stats section on NBA.com shows per-game hustle numbers and leaders, which you can scrape or manually collect if needed.
   For a dataset you can download and analyze easily, use one of the API wrappers to request hustle stats per game and export to CSV/JSON.